Worksheet 1.5 — Derivative Using Definition

Grade 12 Calculus and Vectors (MCV4U) · Unit 1: Rates of Change and Limits

The Big Ideas

$f'(x) = \lim_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$, with x left general (unlike Lesson 1.4's single point). Expand, simplify to cancel h, then let $h \to 0$.

- $f'(x) = \lim_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$.
- Layout: Step 1 — find $f(x+h)$ and expand it. Step 2 — substitute into the definition, cross out the terms that cancel, factor out h, cancel it, and box the answer.
- Combine fractions / rationalize so h cancels.
- Once $f'(x)$ is known, the slope at any point a is just $f'(a)$ — no limit needed.

The Big Picture

The definition

The **derivative** is $f'(x) = \lim_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$ the same limit as Lesson 1.4, but with x left general instead of one fixed point.

What it measures

$f'(x)$ gives the instantaneous rate of change, equivalently the slope of the tangent line, at *every* x at once.

Using the definition

Computing this limit directly is differentiating *using the definition* (or *first principles*), and it justifies the shortcut rules that follow.

Worked Examples

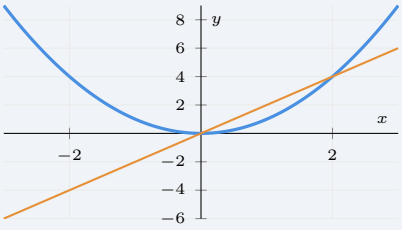
Example 1 — Polynomial

Differentiate $f(x) = x^2$ using the definition.

Solution. **Step 1:** Find $f(x+h)$: $f(x+h) = (x+h)^2 = x^2 + 2xh + h^2$.

Step 2: Substitute into the definition, cross out what cancels, factor out h, and cancel it:

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h\to 0} \frac{(x^2 + 2xh + h^2) - (x)^2}{h} \\ &= \lim_{h\to 0} \frac{\cancel{x^2} + 2xh + h^2 - \cancel{x^2}}{h} \\ &= \lim_{h\to 0} \frac{\cancel{h}(2x+h)}{\cancel{h}} \\ &= \boxed{2x} \end{aligned}$$



Example 2 — Polynomial

Differentiate $f(x) = 3x^2 - x$ using the definition.

Solution. **Step 1:** Find $f(x+h)$: $f(x+h) = 3(x+h)^2 - (x+h) = 3x^2 + 6xh + 3h^2 - x - h$.

Step 2: Substitute into the definition, cross out what cancels, factor out h, and cancel it:

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h\to 0} \frac{(3x^2 + 6xh + 3h^2 - x - h) - (3x^2 - x)}{h} \\ &= \lim_{h\to 0} \frac{\cancel{3x^2} + 6xh + 3h^2 - \cancel{x} - h - \cancel{3x^2} + \cancel{x}}{h} \\ &= \lim_{h\to 0} \frac{\cancel{h}(6x + 3h - 1)}{\cancel{h}} \\ &= \boxed{6x - 1} \end{aligned}$$

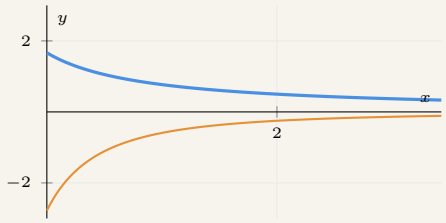
Example 3 — Reciprocal

Differentiate $f(x) = \dfrac{1}{x}$ using the definition.

Solution. **Step 1:** Find $f(x+h)$: $f(x+h) = \dfrac{1}{x+h}$.

Step 2: Substitute into the definition, combine the fractions over the common denominator $x(x+h)$, cross out what cancels, and cancel h:

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} \\ &= \lim_{h\to 0} \frac{\dfrac{1}{x+h} - \dfrac{1}{x}}{h} \\ &= \lim_{h\to 0} \frac{\dfrac{x-(x+h)}{x(x+h)}}{h} \\ &= \lim_{h\to 0} \frac{x - x - h}{x(x+h)\cdot h} \\ &= \lim_{h\to 0} \frac{-h}{x(x+h)h} \\ &= \lim_{h\to 0} \frac{-1}{x(x+h)} \\ &= \boxed{-\frac{1}{x^2}} \end{aligned}$$



Example 4 — Using the formula

From Example 1, $f'(x) = 2x$. Find the tangent slope of $f(x) = x^2$ at $x = 5$.

Solution. **Step 1:** We already have $f'(x) = 2x$ as a formula valid for every x — no need to redo the limit.

Step 2 (substitute): $f'(5) = 2(5)$.

Conclusion: 10 — instant, once $f'(x)$ is known (compare to redoing the limit in Lesson 1.4).

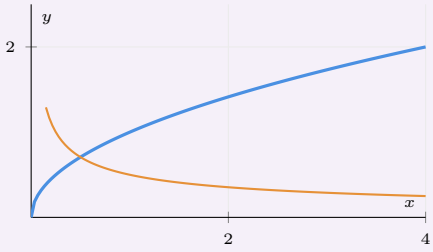
Example 5 — Radical

Differentiate $f(x) = \sqrt{x}$ using the definition.

Solution. **Step 1:** Find $f(x+h)$: $f(x+h) = \sqrt{x+h}$.

Step 2: Substitute into the definition, multiply top and bottom by the conjugate $\sqrt{x+h}+\sqrt{x}$, cross out what cancels, and cancel h:

$$\begin{aligned}
f'(x) &= \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} \\
&= \lim_{h\to 0} \frac{\sqrt{x+h}-\sqrt{x}}{h} \\
&= \lim_{h\to 0} \frac{\left(\sqrt{x+h}-\sqrt{x}\right)\left(\sqrt{x+h}+\sqrt{x}\right)}{h\left(\sqrt{x+h}+\sqrt{x}\right)} \\
&= \lim_{h\to 0} \frac{\cancel{x}+h-\cancel{x}}{h\left(\sqrt{x+h}+\sqrt{x}\right)} \\
&= \lim_{h\to 0} \frac{\cancel{h}}{\cancel{h}\left(\sqrt{x+h}+\sqrt{x}\right)} \\
&= \lim_{h\to 0} \frac{1}{\sqrt{x+h}+\sqrt{x}} \\
&= \boxed{\frac{1}{2\sqrt{x}}}
\end{aligned}$$



Example 6 — Polynomial

Differentiate $f(x) = x^2 + 5x$ using the definition.

Solution. **Step 1:** Find $f(x+h)$: $f(x+h) = (x+h)^2 + 5(x+h) = x^2 + 2xh + h^2 + 5x + 5h$.

Step 2: Substitute into the definition, cross out what cancels, factor out h, and cancel it:

$$\begin{aligned}
f'(x) &= \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} \\
&= \lim_{h\to 0} \frac{(x^2+2xh+h^2+5x+5h)-(x^2+5x)}{h} \\
&= \lim_{h\to 0} \frac{\cancel{x^2}+2xh+h^2+\cancel{5x}+5h-\cancel{x^2}-\cancel{5x}}{h} \\
&= \lim_{h\to 0} \frac{\cancel{h}(2x+h+5)}{\cancel{h}} \\
&= \boxed{2x+5}
\end{aligned}$$

Example 7 — Polynomial

Differentiate $f(x) = 2x^2$ using the definition.

Solution. **Step 1:** Find $f(x+h)$: $f(x+h) = 2(x+h)^2 = 2x^2 + 4xh + 2h^2$.

Step 2: Substitute into the definition, cross out what cancels, factor out h, and cancel it:

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h\to 0} \frac{(2x^2 + 4xh + 2h^2) - (2x^2)}{h} \\ &= \lim_{h\to 0} \frac{\cancel{2x^2} + 4xh + 2h^2 - \cancel{2x^2}}{h} \\ &= \lim_{h\to 0} \frac{\cancel{h}(4x + 2h)}{\cancel{h}} \\ &= \boxed{4x} \end{aligned}$$



Example 8 — Using the formula

From Example 7, $f'(x) = 4x$ for $f(x) = 2x^2$. Find the tangent slope at $x = 1$.

Solution. **Step 1:** Substitute directly into the formula $f'(x) = 4x$.

Step 2: $f'(1) = 4(1)$.

Conclusion: 4 — no limit needed.

Example 9 — Linear

Differentiate $f(x) = 5x - 3$ using the definition.

Solution. **Step 1:** Find $f(x+h)$: $f(x+h) = 5(x+h) - 3 = 5x + 5h - 3$.

Step 2: Substitute into the definition, cross out what cancels, and cancel h:

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h\to 0} \frac{(5x + 5h - 3) - (5x - 3)}{h} \\ &= \lim_{h\to 0} \frac{\cancel{5x} + 5h - \cancel{3} - \cancel{5x} + \cancel{3}}{h} \\ &= \lim_{h\to 0} \frac{5\cancel{h}}{\cancel{h}} \\ &= \boxed{5} \end{aligned}$$

A line's derivative is its constant slope.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Using the definition, differentiate $f(x) = x^2$.

Question 2

Using the definition, differentiate $f(x) = x^2 + 5x$.

Question 3

Using the definition, differentiate $f(x) = 2x^2$.

Question 4

Given $f'(x) = 2x$ for $f(x) = x^2$, find the tangent slope at $x = 1$.

Question 5

Using the definition, differentiate $f(x) = 5x - 3$.

Question 6

Using the definition, differentiate $f(x) = 4x^2 - x$.

Question 7

Using the definition, differentiate $f(x) = \frac{1}{x}$.

Question 8

Given $f'(x) = 2x$ for $f(x) = x^2$, find the tangent slope at $x = 2$.

Question 9

Using the definition, differentiate $f(x) = x^2 - 3x$.

Question 10

Using the definition, differentiate $f(x) = 7$.

Question 11

Given $f'(x) = 2x$ for $f(x) = x^2$, find the tangent slope at $x = -1$.

Question 12

Using the definition, differentiate $f(x) = 3x^2$.

Question 13 — Challenge

Using the definition, differentiate $f(x) = \sqrt{x}$ (rationalize).

Question Recap & Answer Key

Question Recap

1. Using the definition, differentiate $f(x) = x^2$.
2. Using the definition, differentiate $f(x) = x^2 + 5x$.
3. Using the definition, differentiate $f(x) = 2x^2$.
4. Given $f'(x) = 2x$ for $f(x) = x^2$, find the tangent slope at $x = 1$.
5. Using the definition, differentiate $f(x) = 5x - 3$.
6. Using the definition, differentiate $f(x) = 4x^2 - x$.
7. Using the definition, differentiate $f(x) = \dfrac{1}{x}$.
8. Given $f'(x) = 2x$ for $f(x) = x^2$, find the tangent slope at $x = 2$.
9. Using the definition, differentiate $f(x) = x^2 - 3x$.
10. Using the definition, differentiate $f(x) = 7$.
11. Given $f'(x) = 2x$ for $f(x) = x^2$, find the tangent slope at $x = -1$.
12. Using the definition, differentiate $f(x) = 3x^2$.
13. Using the definition, differentiate $f(x) = \sqrt{x}$ (rationalize).

✓ Answer Key

1. $2x$
2. $2x + 5$
3. $4x$
4. 2
5. 5
6. $8x - 1$
7. $-\frac{1}{x^2}$
8. 4
9. $2x - 3$
10. 0
11. -2
12. $6x$
13. $\frac{1}{2\sqrt{x}}$